

18.1 部分习题参考解答

$$1. \quad (1) \quad f(x) = \sum_{n=0}^{\infty} \frac{4\cos(2n+1)x}{(2n+1)^2\pi}, \quad x \in [-\pi, \pi];$$

$$(2) \quad \pi + \frac{4}{3}\pi^2 + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos(nx) - \frac{2+4\pi}{n} \sin(nx) = \begin{cases} x+x^2, & x \in (0, 2\pi), \\ \pi+2\pi^2, & x=0, 2\pi. \end{cases}$$

$$(3) \quad \frac{1}{2} - \frac{\pi}{4} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2} - \sum_{n=1}^{\infty} \frac{(-1)^n(1+\pi)-1}{n\pi} \sin nx = \begin{cases} 1, & -\pi < x < 0, \\ -x, & 0 < x < \pi \\ \frac{1}{2}, & x=0, \\ \frac{1-\pi}{2}, & x=\pm\pi \end{cases}$$

$$(4) \quad \frac{2}{3}\pi^2 + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos nx = \pi^2 - x^2 \quad (-\pi \leq x \leq \pi).$$

$$2. \quad (1) \quad 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin 2nx = \begin{cases} x-1, & x \in (0, \pi), \\ 0, & x=0, \pi. \end{cases}$$

$$(2) \quad \sum_{n=1}^{\infty} \frac{\sin 2nx}{2n} = \begin{cases} \frac{\pi}{4} - \frac{x}{2}, & 0 < x < \pi \\ 0, & x=0, \pi. \end{cases}$$

$$(3) \quad \frac{h}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin nh}{n} \cos nx = \begin{cases} 1, & 0 \leq x < h, \\ 0, & h < x \leq \pi, \\ \frac{1}{2}, & x=h \end{cases}$$

$$3. \quad e^x = \frac{e^\pi - e^{-\pi}}{2\pi} + \frac{e^\pi - e^{-\pi}}{\pi} \sum_{n=1}^{\infty} \left[\frac{(-1)^n}{1+n^2} \cos nx + \frac{(-1)^{n-1}n}{n^2+1} \sin nx \right] \quad (-\pi < x < \pi),$$

$$\sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{\pi-1}{2} + \frac{\pi}{e^{2\pi}-1}.$$

$$4. \quad f(x) = \frac{8}{\pi} \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^3} \sin(2n-1)x \quad (\forall x \in [0, \pi]); \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{(2n-1)^3} = \frac{\pi^3}{32}.$$

$$5. \quad (1) \quad \frac{l}{2} - \frac{4l}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi}{l} x = |x| \quad (-l \leq x \leq l);$$

$$(2) \quad \frac{h}{2} + \frac{2h}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin \frac{(2n-1)\pi}{2} x = \begin{cases} 0, & -2 < x < 0, \\ h, & 0 < x < 2, \\ \frac{h}{2}, & x = 0, \pm 2 \end{cases}$$

$$6. \quad f(x) = 1 - \frac{\pi^2}{3} + 4 \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{n^2} \cos nx \quad (\forall x \in [0, \pi]); \quad \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}.$$

$$7. \quad f(x) = \frac{4l}{\pi^2} \sum_{k=0}^{+\infty} \frac{(-1)^k}{(2k+1)^2} \sin \frac{(2k+1)\pi x}{l} \quad (\forall x \in (0, l)).$$

$$8. \quad f(x) \sim \frac{3\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{+\infty} \frac{\cos(2n-1)x}{(2n-1)^2}; \quad \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

$$9. \quad S(-1) = 1.$$

$$10. \quad |\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{+\infty} \frac{\cos 2nx}{4n^2 - 1}, \quad \forall x \in \mathbb{R}.$$

$$11. \quad \bar{a}_n = a_n \cos nh + b_n \sin nh, \quad \bar{b}_n = b_n \cos nh - a_n \sin nh$$

$$12. \quad A_n = \begin{cases} a_n & n = 0 \\ a_n \frac{\sin nh}{nh} & n \neq 0, \end{cases} \quad B_n = b_n \frac{\sin nh}{nh}.$$